

week_11_wederstrandt_d

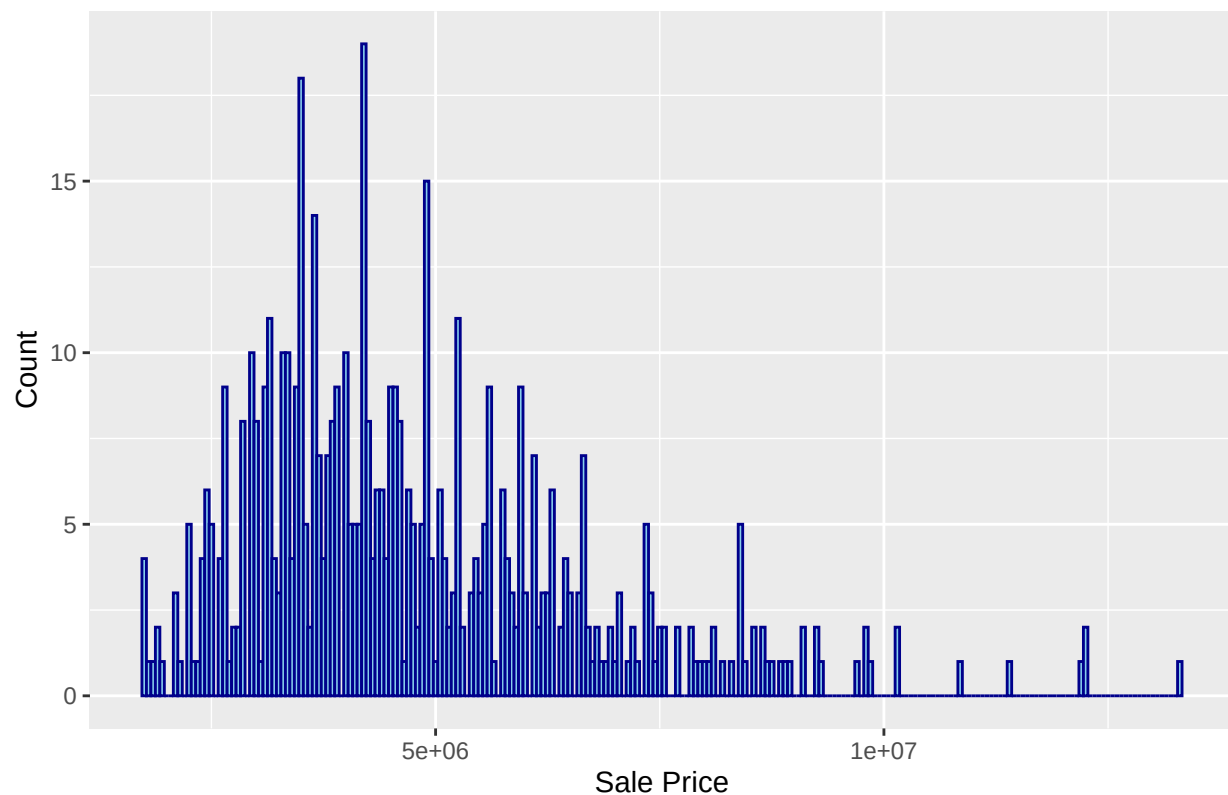
David Wederstrandt

2026-06-28

```
knitr::opts_chunk$set(echo=FALSE)
```

Housing Prices

Histogram of house prices



```
## <theme> List of 144
## $ line : <ggplot2::element_line>
## ..@ colour : chr "black"
## ..@ linewidth : num 0.5
## ..@ linetype : num 1
## ..@ lineend : chr "butt"
## ..@ linejoin : chr "round"
## ..@ arrow : logi FALSE
## ..@ arrow.fill : chr "black"
## ..@ inherit.blank: logi TRUE
## $ rect : <ggplot2::element_rect>
```

```

## ..@ fill      : chr "white"
## ..@ colour    : chr "black"
## ..@ linewidth : num 0.5
## ..@ linetype  : num 1
## ..@ linejoin  : chr "round"
## ..@ inherit.blank: logi TRUE
## $ text                : <ggplot2::element_text>
## ..@ family      : chr ""
## ..@ face        : chr "plain"
## ..@ italic      : chr NA
## ..@ fontweight  : num NA
## ..@ fontwidth   : num NA
## ..@ colour      : chr "black"
## ..@ size        : num 11
## ..@ hjust       : num 0.5
## ..@ vjust       : num 0.5
## ..@ angle       : num 0
## ..@ lineheight  : num 0.9
## ..@ margin      : <ggplot2::margin> num [1:4] 0 0 0 0
## ..@ debug       : logi FALSE
## ..@ inherit.blank: logi TRUE
## $ title                : <ggplot2::element_text>
## ..@ family      : NULL
## ..@ face        : NULL
## ..@ italic      : chr NA
## ..@ fontweight  : num NA
## ..@ fontwidth   : num NA
## ..@ colour      : NULL
## ..@ size        : NULL
## ..@ hjust       : NULL
## ..@ vjust       : NULL
## ..@ angle       : NULL
## ..@ lineheight  : NULL
## ..@ margin      : NULL
## ..@ debug       : NULL
## ..@ inherit.blank: logi TRUE
## $ point                : <ggplot2::element_point>
## ..@ colour      : chr "black"
## ..@ shape       : num 19
## ..@ size        : num 1.5
## ..@ fill        : chr "white"
## ..@ stroke      : num 0.5
## ..@ inherit.blank: logi TRUE
## $ polygon              : <ggplot2::element_polygon>
## ..@ fill        : chr "white"
## ..@ colour      : chr "black"
## ..@ linewidth   : num 0.5
## ..@ linetype    : num 1
## ..@ linejoin    : chr "round"
## ..@ inherit.blank: logi TRUE
## $ geom                  : <ggplot2::element_geom>
## ..@ ink         : chr "black"
## ..@ paper       : chr "white"
## ..@ accent      : chr "#3366FF"

```

```

## ..@ linewidth : num 0.5
## ..@ borderwidth: num 0.5
## ..@ linetype : int 1
## ..@ bordertype : int 1
## ..@ family : chr ""
## ..@ fontsize : num 3.87
## ..@ pointsize : num 1.5
## ..@ pointshape : num 19
## ..@ colour : NULL
## ..@ fill : NULL
## $ spacing : 'simpleUnit' num 5.5points
## ..- attr(*, "unit")= int 8
## $ margins : <ggplot2::margin> num [1:4] 5.5 5.5 5.5 5.5
## $ aspect.ratio : NULL
## $ axis.title : NULL
## $ axis.title.x : <ggplot2::element_text>
## ..@ family : NULL
## ..@ face : NULL
## ..@ italic : chr NA
## ..@ fontweight : num NA
## ..@ fontwidth : num NA
## ..@ colour : NULL
## ..@ size : NULL
## ..@ hjust : NULL
## ..@ vjust : num 1
## ..@ angle : NULL
## ..@ lineheight : NULL
## ..@ margin : <ggplot2::margin> num [1:4] 2.75 0 0 0
## ..@ debug : NULL
## ..@ inherit.blank: logi TRUE
## $ axis.title.x.top : <ggplot2::element_text>
## ..@ family : NULL
## ..@ face : NULL
## ..@ italic : chr NA
## ..@ fontweight : num NA
## ..@ fontwidth : num NA
## ..@ colour : NULL
## ..@ size : NULL
## ..@ hjust : NULL
## ..@ vjust : num 0
## ..@ angle : NULL
## ..@ lineheight : NULL
## ..@ margin : <ggplot2::margin> num [1:4] 0 0 2.75 0
## ..@ debug : NULL
## ..@ inherit.blank: logi TRUE
## $ axis.title.x.bottom : NULL
## $ axis.title.y : <ggplot2::element_text>
## ..@ family : NULL
## ..@ face : NULL
## ..@ italic : chr NA
## ..@ fontweight : num NA
## ..@ fontwidth : num NA
## ..@ colour : NULL
## ..@ size : NULL

```

```

## ..@ hjust      : NULL
## ..@ vjust      : num 1
## ..@ angle      : num 90
## ..@ lineheight : NULL
## ..@ margin     : <ggplot2::margin> num [1:4] 0 2.75 0 0
## ..@ debug      : NULL
## ..@ inherit.blank: logi TRUE
## $ axis.title.y.left      : NULL
## $ axis.title.y.right     : <ggplot2::element_text>
## ..@ family      : NULL
## ..@ face        : NULL
## ..@ italic      : chr NA
## ..@ fontweight  : num NA
## ..@ fontwidth   : num NA
## ..@ colour      : NULL
## ..@ size        : NULL
## ..@ hjust       : NULL
## ..@ vjust       : num 1
## ..@ angle       : num -90
## ..@ lineheight  : NULL
## ..@ margin      : <ggplot2::margin> num [1:4] 0 0 0 2.75
## ..@ debug       : NULL
## ..@ inherit.blank: logi TRUE
## $ axis.text              : <ggplot2::element_text>
## ..@ family        : NULL
## ..@ face          : NULL
## ..@ italic        : chr NA
## ..@ fontweight    : num NA
## ..@ fontwidth     : num NA
## ..@ colour        : chr "#4D4D4DFF"
## ..@ size          : 'rel' num 0.8
## ..@ hjust         : NULL
## ..@ vjust         : NULL
## ..@ angle         : NULL
## ..@ lineheight    : NULL
## ..@ margin        : NULL
## ..@ debug         : NULL
## ..@ inherit.blank: logi TRUE
## $ axis.text.x            : <ggplot2::element_text>
## ..@ family        : NULL
## ..@ face          : NULL
## ..@ italic        : chr NA
## ..@ fontweight    : num NA
## ..@ fontwidth     : num NA
## ..@ colour        : NULL
## ..@ size          : NULL
## ..@ hjust         : NULL
## ..@ vjust         : num 1
## ..@ angle         : NULL
## ..@ lineheight    : NULL
## ..@ margin        : <ggplot2::margin> num [1:4] 2.2 0 0 0
## ..@ debug         : NULL
## ..@ inherit.blank: logi TRUE
## $ axis.text.x.top        : <ggplot2::element_text>

```

```

## ..@ family      : NULL
## ..@ face        : NULL
## ..@ italic      : chr NA
## ..@ fontweight  : num NA
## ..@ fontwidth   : num NA
## ..@ colour      : NULL
## ..@ size        : NULL
## ..@ hjust       : NULL
## ..@ vjust       : NULL
## ..@ angle       : NULL
## ..@ lineheight  : NULL
## ..@ margin      : <ggplot2::margin> num [1:4] 0 0 4.95 0
## ..@ debug       : NULL
## ..@ inherit.blank: logi TRUE
## $ axis.text.x.bottom      : <ggplot2::element_text>
## ..@ family      : NULL
## ..@ face        : NULL
## ..@ italic      : chr NA
## ..@ fontweight  : num NA
## ..@ fontwidth   : num NA
## ..@ colour      : NULL
## ..@ size        : NULL
## ..@ hjust       : NULL
## ..@ vjust       : NULL
## ..@ angle       : NULL
## ..@ lineheight  : NULL
## ..@ margin      : <ggplot2::margin> num [1:4] 4.95 0 0 0
## ..@ debug       : NULL
## ..@ inherit.blank: logi TRUE
## $ axis.text.y      : <ggplot2::element_text>
## ..@ family      : NULL
## ..@ face        : NULL
## ..@ italic      : chr NA
## ..@ fontweight  : num NA
## ..@ fontwidth   : num NA
## ..@ colour      : NULL
## ..@ size        : NULL
## ..@ hjust       : num 1
## ..@ vjust       : NULL
## ..@ angle       : NULL
## ..@ lineheight  : NULL
## ..@ margin      : <ggplot2::margin> num [1:4] 0 2.2 0 0
## ..@ debug       : NULL
## ..@ inherit.blank: logi TRUE
## $ axis.text.y.left  : <ggplot2::element_text>
## ..@ family      : NULL
## ..@ face        : NULL
## ..@ italic      : chr NA
## ..@ fontweight  : num NA
## ..@ fontwidth   : num NA
## ..@ colour      : NULL
## ..@ size        : NULL
## ..@ hjust       : NULL
## ..@ vjust       : NULL

```

```

## ..@ angle      : NULL
## ..@ lineheight  : NULL
## ..@ margin      : <ggplot2::margin> num [1:4] 0 4.95 0 0
## ..@ debug       : NULL
## ..@ inherit.blank: logi TRUE
## $ axis.text.y.right : <ggplot2::element_text>
## ..@ family       : NULL
## ..@ face          : NULL
## ..@ italic        : chr NA
## ..@ fontweight    : num NA
## ..@ fontwidth     : num NA
## ..@ colour        : NULL
## ..@ size          : NULL
## ..@ hjust         : NULL
## ..@ vjust         : NULL
## ..@ angle         : NULL
## ..@ lineheight    : NULL
## ..@ margin        : <ggplot2::margin> num [1:4] 0 0 0 4.95
## ..@ debug         : NULL
## ..@ inherit.blank: logi TRUE
## $ axis.text.theta   : NULL
## $ axis.text.r        : <ggplot2::element_text>
## ..@ family         : NULL
## ..@ face            : NULL
## ..@ italic          : chr NA
## ..@ fontweight      : num NA
## ..@ fontwidth       : num NA
## ..@ colour          : NULL
## ..@ size            : NULL
## ..@ hjust           : num 0.5
## ..@ vjust           : NULL
## ..@ angle           : NULL
## ..@ lineheight      : NULL
## ..@ margin          : <ggplot2::margin> num [1:4] 0 2.2 0 2.2
## ..@ debug           : NULL
## ..@ inherit.blank   : logi TRUE
## $ axis.ticks         : <ggplot2::element_blank>
## $ axis.ticks.x        : NULL
## $ axis.ticks.x.top    : NULL
## $ axis.ticks.x.bottom : NULL
## $ axis.ticks.y        : NULL
## $ axis.ticks.y.left   : NULL
## $ axis.ticks.y.right  : NULL
## $ axis.ticks.theta    : NULL
## $ axis.ticks.r        : NULL
## $ axis.minor.ticks.x.top : NULL
## $ axis.minor.ticks.x.bottom : NULL
## $ axis.minor.ticks.y.left : NULL
## $ axis.minor.ticks.y.right : NULL
## $ axis.minor.ticks.theta : NULL
## $ axis.minor.ticks.r : NULL
## $ axis.ticks.length   : 'rel' num 0.5
## $ axis.ticks.length.x : NULL
## $ axis.ticks.length.x.top : NULL

```

```

## $ axis.ticks.length.x.bottom      : NULL
## $ axis.ticks.length.y             : NULL
## $ axis.ticks.length.y.left        : NULL
## $ axis.ticks.length.y.right       : NULL
## $ axis.ticks.length.theta         : NULL
## $ axis.ticks.length.r             : NULL
## $ axis.minor.ticks.length         : 'rel' num 0.75
## $ axis.minor.ticks.length.x       : NULL
## $ axis.minor.ticks.length.x.top   : NULL
## $ axis.minor.ticks.length.x.bottom : NULL
## $ axis.minor.ticks.length.y       : NULL
## $ axis.minor.ticks.length.y.left  : NULL
## $ axis.minor.ticks.length.y.right : NULL
## $ axis.minor.ticks.length.theta   : NULL
## $ axis.minor.ticks.length.r       : NULL
## $ axis.line                       : <ggplot2::element_blank>
## $ axis.line.x                     : NULL
## $ axis.line.x.top                  : NULL
## $ axis.line.x.bottom               : NULL
## $ axis.line.y                      : NULL
## $ axis.line.y.left                 : NULL
## $ axis.line.y.right                : NULL
## $ axis.line.theta                  : NULL
## $ axis.line.r                      : NULL
## $ legend.background                : <ggplot2::element_blank>
## $ legend.margin                    : NULL
## $ legend.spacing                   : 'rel' num 2
## $ legend.spacing.x                 : NULL
## $ legend.spacing.y                 : NULL
## $ legend.key                       : <ggplot2::element_blank>
## $ legend.key.size                   : 'simpleUnit' num 1.2lines
##   .- attr(*, "unit")= int 3
## $ legend.key.height                 : NULL
## $ legend.key.width                  : NULL
## $ legend.key.spacing                : NULL
## $ legend.key.spacing.x              : NULL
## $ legend.key.spacing.y              : NULL
## $ legend.key.justification          : NULL
## $ legend.frame                      : NULL
## $ legend.ticks                     : NULL
## $ legend.ticks.length               : 'rel' num 0.2
## $ legend.axis.line                  : NULL
## $ legend.text                       : <ggplot2::element_text>
##   .@ family       : NULL
##   .@ face         : NULL
##   .@ italic       : chr NA
##   .@ fontweight   : num NA
##   .@ fontwidth    : num NA
##   .@ colour       : NULL
##   .@ size         : 'rel' num 0.8
##   .@ hjust        : NULL
##   .@ vjust        : NULL
##   .@ angle        : NULL
##   .@ lineheight   : NULL

```

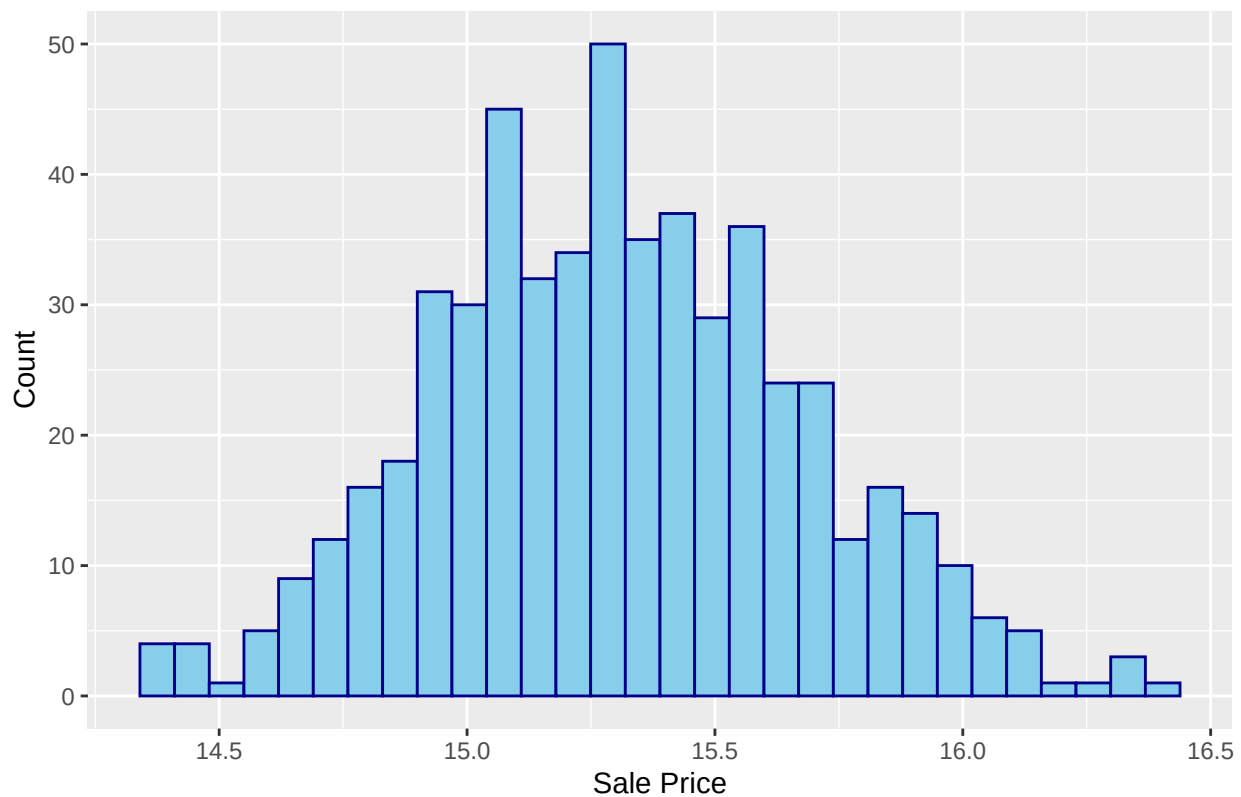
```
## ..@ margin      : NULL
## ..@ debug       : NULL
## ..@ inherit.blank: logi TRUE
## $ legend.text.position : NULL
## $ legend.title      : <ggplot2::element_text>
## ..@ family       : NULL
## ..@ face         : NULL
## ..@ italic       : chr NA
## ..@ fontweight   : num NA
## ..@ fontwidth    : num NA
## ..@ colour       : NULL
## ..@ size         : NULL
## ..@ hjust        : num 0
## ..@ vjust        : NULL
## ..@ angle        : NULL
## ..@ lineheight   : NULL
## ..@ margin       : NULL
## ..@ debug        : NULL
## ..@ inherit.blank: logi TRUE
## $ legend.title.position : NULL
## $ legend.position      : chr "right"
## $ legend.position.inside : NULL
## $ legend.direction     : NULL
## $ legend.byrow         : NULL
## $ legend.justification : chr "center"
## $ legend.justification.top : NULL
## $ legend.justification.bottom : NULL
## $ legend.justification.left : NULL
## $ legend.justification.right : NULL
## $ legend.justification.inside : NULL
## [list output truncated]
## @ complete: logi TRUE
## @ validate: logi TRUE
```

I can see the price is right skewed, which can make it difficult to make assumptions about.

Explain any transformations or modifications you made to the dataset.

The price data is right-skewed so I will transform this column I will take the log of price to create a normal(ish) distribution.

Histogram of house prices



```
## <theme> List of 144
## $ line : <ggplot2::element_line>
## ..@ colour      : chr "black"
## ..@ linewidth   : num 0.5
## ..@ linetype    : num 1
## ..@ lineend     : chr "butt"
## ..@ linejoin    : chr "round"
## ..@ arrow       : logi FALSE
## ..@ arrow.fill  : chr "black"
## ..@ inherit.blank: logi TRUE
## $ rect : <ggplot2::element_rect>
## ..@ fill       : chr "white"
## ..@ colour     : chr "black"
## ..@ linewidth  : num 0.5
## ..@ linetype   : num 1
## ..@ linejoin   : chr "round"
## ..@ inherit.blank: logi TRUE
## $ text : <ggplot2::element_text>
## ..@ family     : chr ""
## ..@ face       : chr "plain"
## ..@ italic     : chr NA
## ..@ fontweight : num NA
## ..@ fontwidth  : num NA
## ..@ colour     : chr "black"
## ..@ size       : num 11
## ..@ hjust      : num 0.5
## ..@ vjust      : num 0.5
```

```

## ..@ angle      : num 0
## ..@ lineheight  : num 0.9
## ..@ margin      : <ggplot2::margin> num [1:4] 0 0 0 0
## ..@ debug       : logi FALSE
## ..@ inherit.blank: logi TRUE
## $ title          : <ggplot2::element_text>
## ..@ family      : NULL
## ..@ face        : NULL
## ..@ italic      : chr NA
## ..@ fontweight  : num NA
## ..@ fontwidth   : num NA
## ..@ colour      : NULL
## ..@ size        : NULL
## ..@ hjust       : NULL
## ..@ vjust       : NULL
## ..@ angle       : NULL
## ..@ lineheight  : NULL
## ..@ margin      : NULL
## ..@ debug       : NULL
## ..@ inherit.blank: logi TRUE
## $ point          : <ggplot2::element_point>
## ..@ colour      : chr "black"
## ..@ shape       : num 19
## ..@ size        : num 1.5
## ..@ fill        : chr "white"
## ..@ stroke      : num 0.5
## ..@ inherit.blank: logi TRUE
## $ polygon        : <ggplot2::element_polygon>
## ..@ fill        : chr "white"
## ..@ colour      : chr "black"
## ..@ linewidth   : num 0.5
## ..@ linetype    : num 1
## ..@ linejoin    : chr "round"
## ..@ inherit.blank: logi TRUE
## $ geom           : <ggplot2::element_geom>
## ..@ ink         : chr "black"
## ..@ paper       : chr "white"
## ..@ accent      : chr "#3366FF"
## ..@ linewidth   : num 0.5
## ..@ borderwidth : num 0.5
## ..@ linetype    : int 1
## ..@ bordertype  : int 1
## ..@ family      : chr ""
## ..@ fontsize    : num 3.87
## ..@ pointsize   : num 1.5
## ..@ pointshape  : num 19
## ..@ colour      : NULL
## ..@ fill        : NULL
## $ spacing        : 'simpleUnit' num 5.5points
## ..- attr(*, "unit")= int 8
## $ margins         : <ggplot2::margin> num [1:4] 5.5 5.5 5.5 5.5
## $ aspect.ratio    : NULL
## $ axis.title       : NULL
## $ axis.title.x     : <ggplot2::element_text>

```

```

## ..@ family      : NULL
## ..@ face        : NULL
## ..@ italic      : chr NA
## ..@ fontweight  : num NA
## ..@ fontwidth   : num NA
## ..@ colour      : NULL
## ..@ size        : NULL
## ..@ hjust       : NULL
## ..@ vjust       : num 1
## ..@ angle       : NULL
## ..@ lineheight  : NULL
## ..@ margin      : <ggplot2::margin> num [1:4] 2.75 0 0 0
## ..@ debug       : NULL
## ..@ inherit.blank: logi TRUE
## $ axis.title.x.top      : <ggplot2::element_text>
## ..@ family      : NULL
## ..@ face        : NULL
## ..@ italic      : chr NA
## ..@ fontweight  : num NA
## ..@ fontwidth   : num NA
## ..@ colour      : NULL
## ..@ size        : NULL
## ..@ hjust       : NULL
## ..@ vjust       : num 0
## ..@ angle       : NULL
## ..@ lineheight  : NULL
## ..@ margin      : <ggplot2::margin> num [1:4] 0 0 2.75 0
## ..@ debug       : NULL
## ..@ inherit.blank: logi TRUE
## $ axis.title.x.bottom   : NULL
## $ axis.title.y          : <ggplot2::element_text>
## ..@ family      : NULL
## ..@ face        : NULL
## ..@ italic      : chr NA
## ..@ fontweight  : num NA
## ..@ fontwidth   : num NA
## ..@ colour      : NULL
## ..@ size        : NULL
## ..@ hjust       : NULL
## ..@ vjust       : num 1
## ..@ angle       : num 90
## ..@ lineheight  : NULL
## ..@ margin      : <ggplot2::margin> num [1:4] 0 2.75 0 0
## ..@ debug       : NULL
## ..@ inherit.blank: logi TRUE
## $ axis.title.y.left     : NULL
## $ axis.title.y.right    : <ggplot2::element_text>
## ..@ family      : NULL
## ..@ face        : NULL
## ..@ italic      : chr NA
## ..@ fontweight  : num NA
## ..@ fontwidth   : num NA
## ..@ colour      : NULL
## ..@ size        : NULL

```

```

## ..@ hjust      : NULL
## ..@ vjust      : num 1
## ..@ angle      : num -90
## ..@ lineheight : NULL
## ..@ margin     : <ggplot2::margin> num [1:4] 0 0 0 2.75
## ..@ debug      : NULL
## ..@ inherit.blank: logi TRUE
## $ axis.text    : <ggplot2::element_text>
## ..@ family     : NULL
## ..@ face       : NULL
## ..@ italic     : chr NA
## ..@ fontweight : num NA
## ..@ fontwidth  : num NA
## ..@ colour     : chr "#4D4D4DFF"
## ..@ size       : 'rel' num 0.8
## ..@ hjust      : NULL
## ..@ vjust      : NULL
## ..@ angle      : NULL
## ..@ lineheight : NULL
## ..@ margin     : NULL
## ..@ debug      : NULL
## ..@ inherit.blank: logi TRUE
## $ axis.text.x  : <ggplot2::element_text>
## ..@ family     : NULL
## ..@ face       : NULL
## ..@ italic     : chr NA
## ..@ fontweight : num NA
## ..@ fontwidth  : num NA
## ..@ colour     : NULL
## ..@ size       : NULL
## ..@ hjust      : NULL
## ..@ vjust      : num 1
## ..@ angle      : NULL
## ..@ lineheight : NULL
## ..@ margin     : <ggplot2::margin> num [1:4] 2.2 0 0 0
## ..@ debug      : NULL
## ..@ inherit.blank: logi TRUE
## $ axis.text.x.top : <ggplot2::element_text>
## ..@ family     : NULL
## ..@ face       : NULL
## ..@ italic     : chr NA
## ..@ fontweight : num NA
## ..@ fontwidth  : num NA
## ..@ colour     : NULL
## ..@ size       : NULL
## ..@ hjust      : NULL
## ..@ vjust      : NULL
## ..@ angle      : NULL
## ..@ lineheight : NULL
## ..@ margin     : <ggplot2::margin> num [1:4] 0 0 4.95 0
## ..@ debug      : NULL
## ..@ inherit.blank: logi TRUE
## $ axis.text.x.bottom : <ggplot2::element_text>
## ..@ family     : NULL

```

```

## ..@ face          : NULL
## ..@ italic        : chr NA
## ..@ fontweight    : num NA
## ..@ fontwidth     : num NA
## ..@ colour        : NULL
## ..@ size          : NULL
## ..@ hjust         : NULL
## ..@ vjust         : NULL
## ..@ angle         : NULL
## ..@ lineheight    : NULL
## ..@ margin        : <ggplot2::margin> num [1:4] 4.95 0 0 0
## ..@ debug         : NULL
## ..@ inherit.blank: logi TRUE
## $ axis.text.y      : <ggplot2::element_text>
## ..@ family        : NULL
## ..@ face          : NULL
## ..@ italic        : chr NA
## ..@ fontweight    : num NA
## ..@ fontwidth     : num NA
## ..@ colour        : NULL
## ..@ size          : NULL
## ..@ hjust         : num 1
## ..@ vjust         : NULL
## ..@ angle         : NULL
## ..@ lineheight    : NULL
## ..@ margin        : <ggplot2::margin> num [1:4] 0 2.2 0 0
## ..@ debug         : NULL
## ..@ inherit.blank: logi TRUE
## $ axis.text.y.left : <ggplot2::element_text>
## ..@ family        : NULL
## ..@ face          : NULL
## ..@ italic        : chr NA
## ..@ fontweight    : num NA
## ..@ fontwidth     : num NA
## ..@ colour        : NULL
## ..@ size          : NULL
## ..@ hjust         : NULL
## ..@ vjust         : NULL
## ..@ angle         : NULL
## ..@ lineheight    : NULL
## ..@ margin        : <ggplot2::margin> num [1:4] 0 4.95 0 0
## ..@ debug         : NULL
## ..@ inherit.blank: logi TRUE
## $ axis.text.y.right : <ggplot2::element_text>
## ..@ family        : NULL
## ..@ face          : NULL
## ..@ italic        : chr NA
## ..@ fontweight    : num NA
## ..@ fontwidth     : num NA
## ..@ colour        : NULL
## ..@ size          : NULL
## ..@ hjust         : NULL
## ..@ vjust         : NULL
## ..@ angle         : NULL

```

```

## ..@ lineheight : NULL
## ..@ margin      : <ggplot2::margin> num [1:4] 0 0 0 4.95
## ..@ debug       : NULL
## ..@ inherit.blank: logi TRUE
## $ axis.text.theta : NULL
## $ axis.text.r      : <ggplot2::element_text>
## ..@ family        : NULL
## ..@ face           : NULL
## ..@ italic         : chr NA
## ..@ fontweight     : num NA
## ..@ fontwidth      : num NA
## ..@ colour         : NULL
## ..@ size           : NULL
## ..@ hjust          : num 0.5
## ..@ vjust          : NULL
## ..@ angle          : NULL
## ..@ lineheight     : NULL
## ..@ margin         : <ggplot2::margin> num [1:4] 0 2.2 0 2.2
## ..@ debug          : NULL
## ..@ inherit.blank: logi TRUE
## $ axis.ticks        : <ggplot2::element_blank>
## $ axis.ticks.x       : NULL
## $ axis.ticks.x.top   : NULL
## $ axis.ticks.x.bottom : NULL
## $ axis.ticks.y       : NULL
## $ axis.ticks.y.left  : NULL
## $ axis.ticks.y.right : NULL
## $ axis.ticks.theta   : NULL
## $ axis.ticks.r       : NULL
## $ axis.minor.ticks.x.top : NULL
## $ axis.minor.ticks.x.bottom : NULL
## $ axis.minor.ticks.y.left : NULL
## $ axis.minor.ticks.y.right : NULL
## $ axis.minor.ticks.theta : NULL
## $ axis.minor.ticks.r : NULL
## $ axis.ticks.length  : 'rel' num 0.5
## $ axis.ticks.length.x : NULL
## $ axis.ticks.length.x.top : NULL
## $ axis.ticks.length.x.bottom : NULL
## $ axis.ticks.length.y : NULL
## $ axis.ticks.length.y.left : NULL
## $ axis.ticks.length.y.right : NULL
## $ axis.ticks.length.theta : NULL
## $ axis.ticks.length.r : NULL
## $ axis.minor.ticks.length : 'rel' num 0.75
## $ axis.minor.ticks.length.x : NULL
## $ axis.minor.ticks.length.x.top : NULL
## $ axis.minor.ticks.length.x.bottom : NULL
## $ axis.minor.ticks.length.y : NULL
## $ axis.minor.ticks.length.y.left : NULL
## $ axis.minor.ticks.length.y.right : NULL
## $ axis.minor.ticks.length.theta : NULL
## $ axis.minor.ticks.length.r : NULL
## $ axis.line          : <ggplot2::element_blank>

```

```

## $ axis.line.x : NULL
## $ axis.line.x.top : NULL
## $ axis.line.x.bottom : NULL
## $ axis.line.y : NULL
## $ axis.line.y.left : NULL
## $ axis.line.y.right : NULL
## $ axis.line.theta : NULL
## $ axis.line.r : NULL
## $ legend.background : <ggplot2::element_blank>
## $ legend.margin : NULL
## $ legend.spacing : 'rel' num 2
## $ legend.spacing.x : NULL
## $ legend.spacing.y : NULL
## $ legend.key : <ggplot2::element_blank>
## $ legend.key.size : 'simpleUnit' num 1.2lines
## ..- attr(*, "unit")= int 3
## $ legend.key.height : NULL
## $ legend.key.width : NULL
## $ legend.key.spacing : NULL
## $ legend.key.spacing.x : NULL
## $ legend.key.spacing.y : NULL
## $ legend.key.justification : NULL
## $ legend.frame : NULL
## $ legend.ticks : NULL
## $ legend.ticks.length : 'rel' num 0.2
## $ legend.axis.line : NULL
## $ legend.text : <ggplot2::element_text>
## ..@ family : NULL
## ..@ face : NULL
## ..@ italic : chr NA
## ..@ fontweight : num NA
## ..@ fontwidth : num NA
## ..@ colour : NULL
## ..@ size : 'rel' num 0.8
## ..@ hjust : NULL
## ..@ vjust : NULL
## ..@ angle : NULL
## ..@ lineheight : NULL
## ..@ margin : NULL
## ..@ debug : NULL
## ..@ inherit.blank: logi TRUE
## $ legend.text.position : NULL
## $ legend.title : <ggplot2::element_text>
## ..@ family : NULL
## ..@ face : NULL
## ..@ italic : chr NA
## ..@ fontweight : num NA
## ..@ fontwidth : num NA
## ..@ colour : NULL
## ..@ size : NULL
## ..@ hjust : num 0
## ..@ vjust : NULL
## ..@ angle : NULL
## ..@ lineheight : NULL

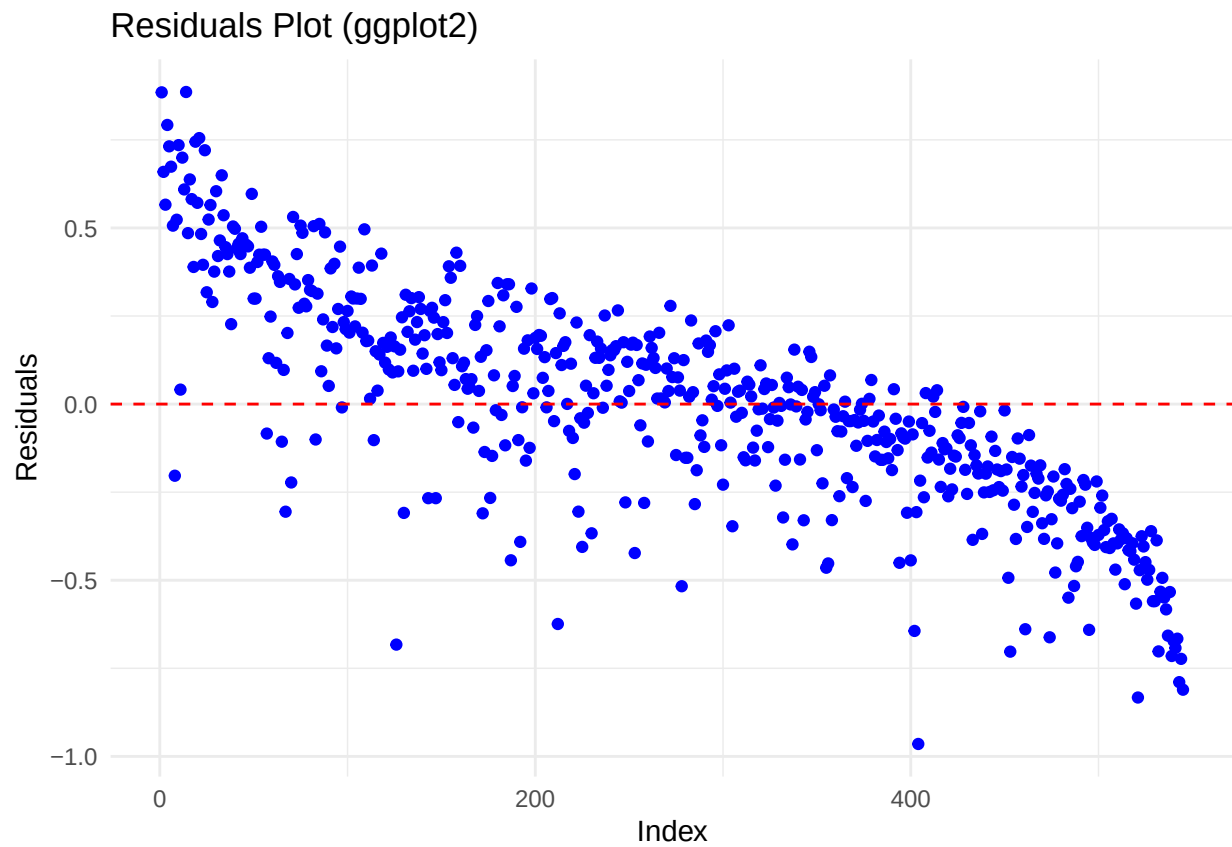
```

```
## ..@ margin      : NULL
## ..@ debug       : NULL
## ..@ inherit.blank: logi TRUE
## $ legend.title.position : NULL
## $ legend.position      : chr "right"
## $ legend.position.inside : NULL
## $ legend.direction     : NULL
## $ legend.byrow         : NULL
## $ legend.justification  : chr "center"
## $ legend.justification.top : NULL
## $ legend.justification.bottom : NULL
## $ legend.justification.left : NULL
## $ legend.justification.right : NULL
## $ legend.justification.inside : NULL
## [list output truncated]
## @ complete: logi TRUE
## @ validate: logi TRUE
```

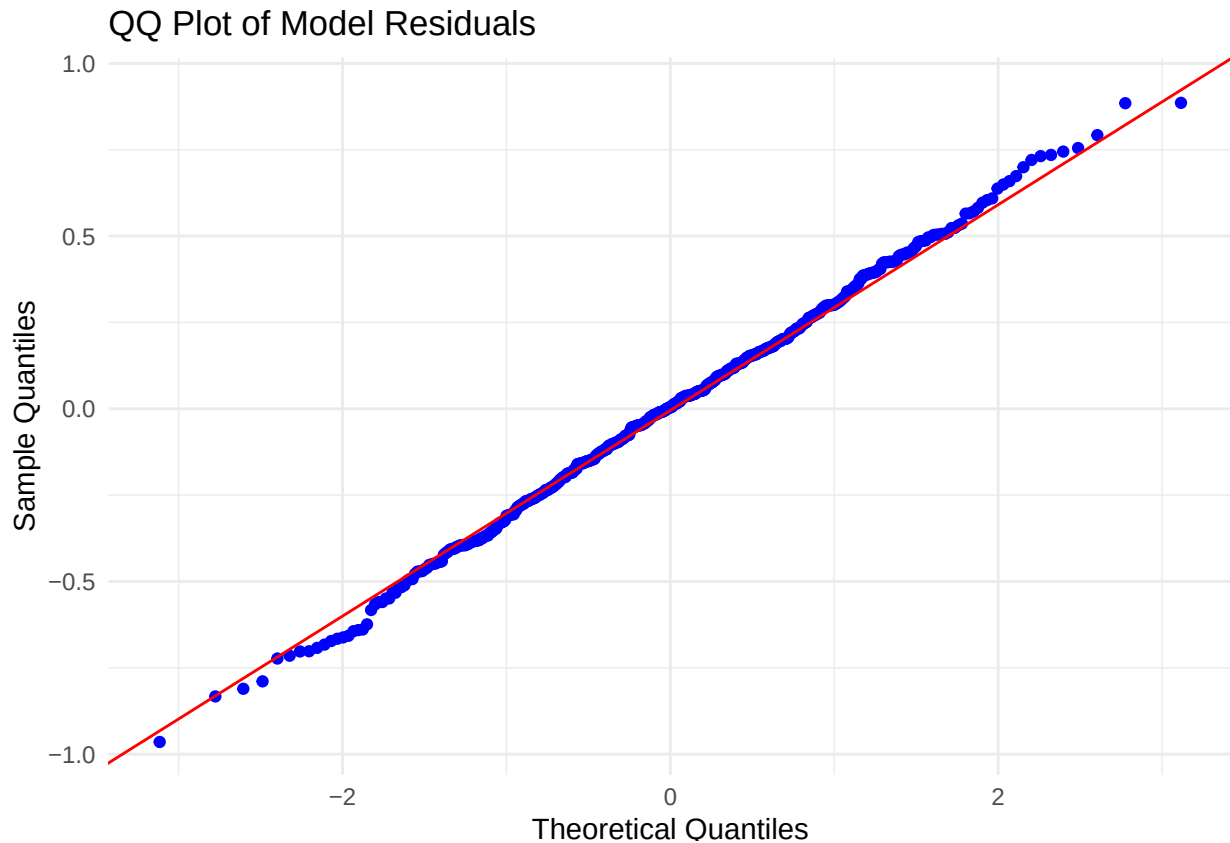
Using the log of price, we now have a normal distribution, but not perfectly normal

Linear Model Results

```
##
## The R-squared of 0.2951017 or 29.5% of the variation in price can be explained by area.
##
##
## The adjusted R-squared of 0.2938035 or 29.3%, tells us that sill only 29% of the variation can be e
## area, even after accounting foe the number of predictors.
##
##
## The RMSE of 0.3121765 tells us that, on average, the predictors differ from observed data.
##
##
## The MSE of 0.09745415 means that, on average, the squared differences between the model's predicted p
## and actual prices are close to 0.09 units squared.
```

You can clearly see a pattern, which suggest that the relationship between area and $\log(\text{price})$ may not be perfectly linear. So, this model would not be appropriate for this data.



Almost all point fall directly on the red line, which tells us the residuals are approximately normal This supports the use of linear regression

Multiple Predictor Linear Regression Model

Multiple Predictors and Reasoning

Category	Reasoning
<i>Area</i>	Most realators values homes at so many dollars per square foot. Bigger houses sell for more money.
<i>Bedrooms</i>	More bathrooms are more appealing to families.
<i>Bathrooms</i>	More bathrooms add convenience, thus value.
<i>stories</i>	More stories require less land to make a larger home.
<i>parking</i>	Small parking areas require more management, while more parking is easier to manage
<i>airconditioning</i>	Depending on location, air conditioning is practically an essential

Multiple Regressor Summary

```
##
## Call:
## lm(formula = Log_SalePrice ~ area + bedrooms + bathrooms + stories +
##     parking + airconditioning, data = housing)
##
```

```
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.71353 -0.15468  0.01021  0.15679  0.74506
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   1.437e+01  4.774e-02 300.980 < 2e-16 ***
## area          6.234e-05  5.187e-06 12.019 < 2e-16 ***
## bedrooms      4.263e-02  1.597e-02  2.670 0.00781 **
## bathrooms     1.866e-01  2.291e-02  8.144 2.68e-15 ***
## stories       8.268e-02  1.367e-02  6.050 2.72e-09 ***
## parking       6.077e-02  1.290e-02  4.710 3.16e-06 ***
## airconditioningyes 1.877e-01  2.374e-02  7.908 1.49e-14 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2395 on 538 degrees of freedom
## Multiple R-squared:  0.5903, Adjusted R-squared:  0.5857
## F-statistic: 129.2 on 6 and 538 DF, p-value: < 2.2e-16
```

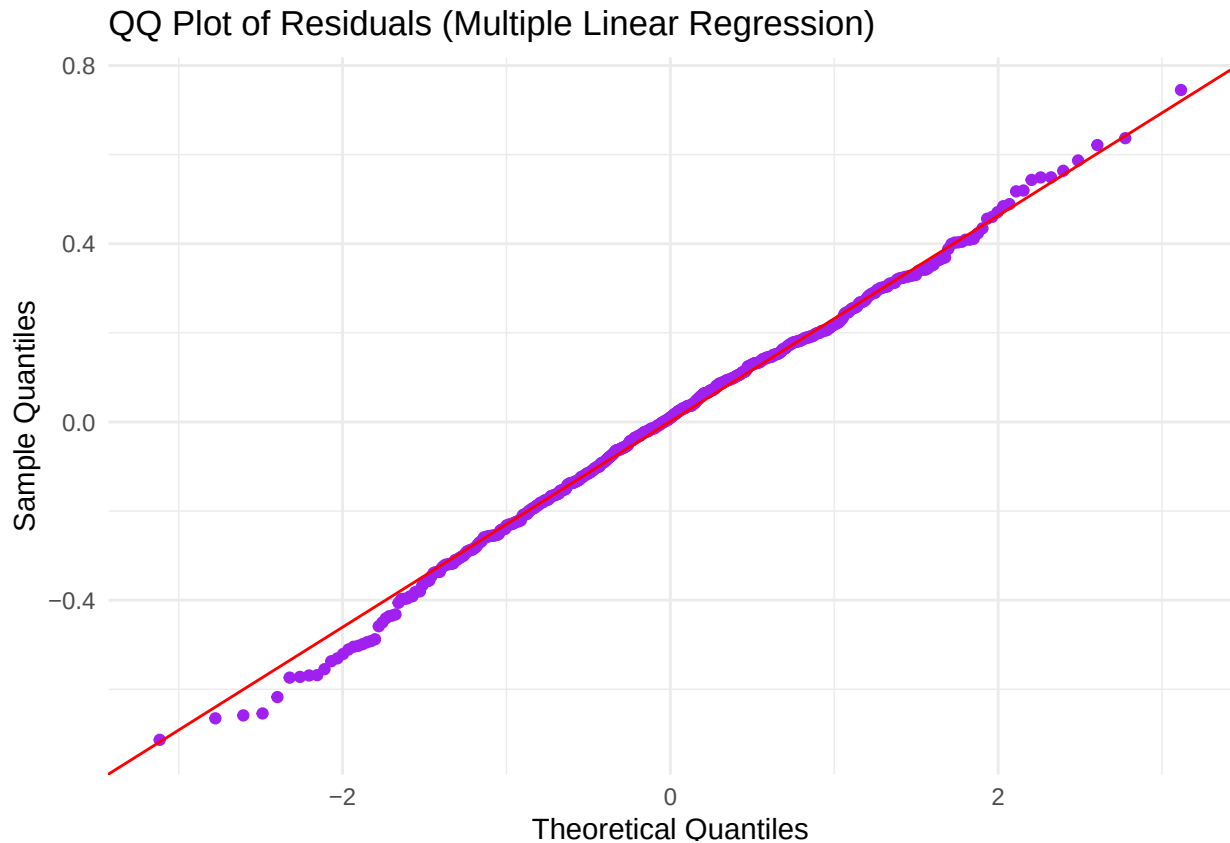
Results

```
##
## The coefficients show the estimated effect of each predictor on log(price), holding other variables constant.
##
## The R-squared and adjusted R-squared values indicate how much of the variation in log(sale price) is explained by the model.
##
## Significant p-values (Pr(>|t|)) for predictors suggest those variables have a statistically significant effect on the response.
##
## Compare these values to the single-predictor model to assess improvement.
```

Second Model' Residuals



You can clearly see a pattern, which suggest that the relationship between area and $\log(\text{price})$ may not be perfectly linear. Just like the residuals for the single residual. So, this model would not be appropriate for this data.



Almost all point fall directly on the red line as they did with just area, which tells us the residuals are approximately normal. This supports the use of linear regression

```
## First Model (area only):
```

```
## R-squared: 0.2951017
```

```
## Adjusted R-squared: 0.2938035
```

```
## RMSE: 0.3121765
```

```
## MSE: 0.09745415
```

```
## Second Model (multiple predictors):
```

```
## R-squared: 0.5902653
```

```
## Adjusted R-squared: 0.5856957
```

```
## RMSE: 0.2380062
```

```
## MSE: 0.05664696
```

```
##
```

```
##
```

```
##
```

```
## Multiple regression model(MRM) R-squared: 0.5902653 is greater than Area regression model (ARM) R-squared: 0.2951017
```

```
##
```

```
## MRM Adjusted R-squared: 0.5856957 is greater than ARM Adjusted R-squared: 0.2938035 .
```

```
##
```

```
## Meaning: MRM explains more of the variation in log(price) than the ARM.
```

```
##
```

```
##
```

```
##
```

```
## Multiple regression model(MRM) Root Mean Squared Error (RMSE) : 0.2380062 is less than Area regression model (ARM) RMSE: 0.3121765
```

```

##
## MRM Mean Squared Error (MSE): 0.05664696 is less than ARML Mean Squared Error (MSE): 0.09745415 .
##
## Meaning: MRM RMSE and MSE are less indicating MRM predictions are, on average, closer to the actual
## Analysis of Variance Table
##
## Model 1: Log_SalePrice ~ area
## Model 2: Log_SalePrice ~ area + bedrooms + bathrooms + stories + parking +
##      airconditioning
##   Res.Df    RSS Df Sum of Sq      F    Pr(>F)
## 1      543 53.113
## 2      538 30.873   5      22.24 77.513 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

## With a p-value of NA 3.779964e-61 we know the Multiple regression model is a significantly better mo
## compared to the area regression model

```